

```
        break

    print('Solar panel heading set to:', direction)
❶ if direction == 'nrth':
    print('Warning: Facing north is inefficient for this panel.')
```

This bug can be hard to detect: the typo in the 'nrth' string ❶ is still syntactically correct Python. The program doesn't crash, and it's easy to overlook the lack of a warning message. But if we used constants and made this same typo, the typo would cause the program to crash because Python would notice that a `NRTH` constant doesn't exist:

```
# Set up constants for each cardinal direction:
NORTH = 'north'
SOUTH = 'south'
EAST = 'east'
WEST = 'west'

while True:
    print('Set solar panel direction:')
    direction = input().lower()
    if direction in (NORTH, SOUTH, EAST, WEST):
        break

    print('Solar panel heading set to:', direction)
❶ if direction == NRTH:
    print('Warning: Facing north is inefficient for this panel.')
```

The `NameError` exception raised by the code line with the `NRTH` typo ❶ makes the bug immediately obvious when you run this program:

```
Set solar panel direction:
west
Solar panel heading set to: west
Traceback (most recent call last):
  File "panelset.py", line 14, in <module>
    if direction == NRTH:
NameError: name 'NRTH' is not defined
```

Magic numbers are a code smell because they don't convey their purpose, making your code less readable, harder to update, and prone to undetectable typos. The solution is to use constant variables instead.